

THISAL ARIYARATNE

Cyber Security Undergraduate

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PROFESSIONAL SUMMARY

Cyber Security undergraduate with a sharp ability to bridge offensive tradecraft and defensive operations - from breaking systems in competitive CTFs to analyzing logs, traffic, and artifacts to understand how they were attacked. Proven track record of top-three finishes across national and inter-university CTF competitions, authoring national-level CTF challenges for HackKAP, and leading a competitive CTF team. Backed by four years of daily Linux and hands-on tooling across WireShark, Nmap, Burp Suite, and Metasploit, and currently architecting AP3X, a Rust-based AI security agent.

TECHNICAL SKILLS

Monitoring & Analysis: WireShark, Tshark, Nmap, RustScan, network traffic analysis, PCAP analysis, log analysis, network monitoring

Incident Investigation: Threat identification, digital forensics, evidence-chain documentation, CTF methodology

Security Tools: Burp Suite, Metasploit, Gobuster, Hashcat, Hydra

Networking & Protocols: TCP/IP, DNS, HTTP/HTTPS, OAuth 2.0, SAML 2.0, JWT

Operating Systems: Kali Linux, Arch Linux, Ubuntu, CachyOS, Windows

Programming & Scripting: Python, Bash, C, C++, C#, Rust

Practice Platforms: Hack The Box, TryHackMe, CTFtime, PicoCTF

Projects

AP3X SOC - AI-Assisted Security Operations Workflow | Rust | Author (In Development)

Building an **AI assisted SOC workflow** integrating **Wazuh SIEM** with a custom Rust terminal agent. **Automated alert triage**, **MITRE ATT&CK** mapping, and threat hunting across a simulated attack environment.

AP3X - AI-Powered Terminal Security Assistant | Rust | Author

Developed a custom **Rust-based terminal AI** agent featuring an interactive TUI, session management, **multi-model selection**, and slash-command workflows. The assistant integrates multiple LLM providers (**OpenAI**, **Claude**) with secure API-key handling, and is built to accelerate Linux operations, scripting, and day-to-day cyber security research from a single command-line interface.

Fortress Cloud Security Assessment - IAM / Authentication Review | Author | Academic report

Conducted a **static security assessment** of an authentication service covering OAuth 2.0, SAML 2.0, and JWT flows. Identified critical protocol weaknesses and mapped realistic exploit chains, then documented the findings, business impact, and prioritized remediation guidance in a structured security report - applying the same investigative and reporting discipline used in real-world security operations.

CTF Challenge Design - HackKAP CTF | Author

Archive Avalanche (Forensics) : used 50,000 nested ZIP files across five layers to force competitors into scripted, automated extraction to find the flag

Red Herring Channel (steganography) : embedded decoy flags across EXIF metadata, PNG chunks, green-channel LSB, and ROT13 trailing data, with the true solve path built around red/blue-channel LSB XOR and a decryption key hidden inside a deliberate filename typo.

Operation Intercept 2025 (Hidden Cargo) - Investigation Write-Up | LinkedIn Publication | HackKAP

Investigated and documented a multi-stage CTF challenge that required structured **analysis of visual, symbolic (Hobo sign symbols)**, and audio-based clues. Applied logical enumeration and systematic investigation techniques to identify hidden indicators and reconstruct a clear evidence chain, then published the full methodology as a write-up for the security community.

EXPERIENCE & LEADERSHIP

Challenge Builder & Member - HackKAP Sri Lanka | 2024 - Present

Designed more than 20+ CTF challenges for national-level competitions in categories like Web Exploitation, Cryptography, Binary Exploitation, Reverse Engineering, Forensics, OSINT, and Steganography. Also, helped run two inter-university CTF events end to end.

Team Leader - Cronoz CTF Team (4 members)

Lead solve strategy, task allocation, and live coordination during competitive events.

Secretary - Ph4ntom Club (Cyber Security Club), CICRA Campus | Aug 2024 - 2025

Organized in-house campus CTF competitions and campus events, delivered Linux workshops on command-line and security tooling.

Vice Secretary - Student Interactive Society, CICRA Campus | Apr 2024 - Feb 2025

Coordinated orientations and monthly events, secured corporate sponsorships and managed communications.

ACHIEVEMENTS

1st Place - CODE BREAKERS, HackKAP CTF 2024

1st Place - FLAG HUNTERS, CICRA Campus CTF 2025

1st Runners-Up - CRONNOZ | ZeroXArena 2026 (CICRA) | **1st Runners-Up** - FLAG HUNTERS, SLIIT CS2

2nd Runners-Up - FLAG HUNTERS, LNBTI | Battle of Multiverse 2026 | **2nd Runners-Up** - CRONNOZ | CryptX 2.0 UOJ

4th Place FLAG HUNTERS | CYPHER 3.0 KDU | **Finalist** - CHRONOS | MEDUSA CTF 2025

EDUCATION

B.Sc. (Hons) Cyber Security (Top-Up) - Wrexham University, UK (pathway with CICRA Campus, Sri Lanka) | Expected Oct 2026

Higher National Diploma in Cyber Security - CICRA Campus (affil. Deakin University) | 2024 - 2026

Foundation in Cyber Security - CICRA Campus | 2024

CERTIFICATIONS

Certified Red Team Analyst (CRTA) - In Progress | **Certified Ethical Hacker (CEH)**, (Coursera Pearson) - In Progress

LANGUAGES

Sinhala | English

REFERENCES

Mr. Hasitha Gunasekara | Senior Windows Engineer | LSEG | Lecturer

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